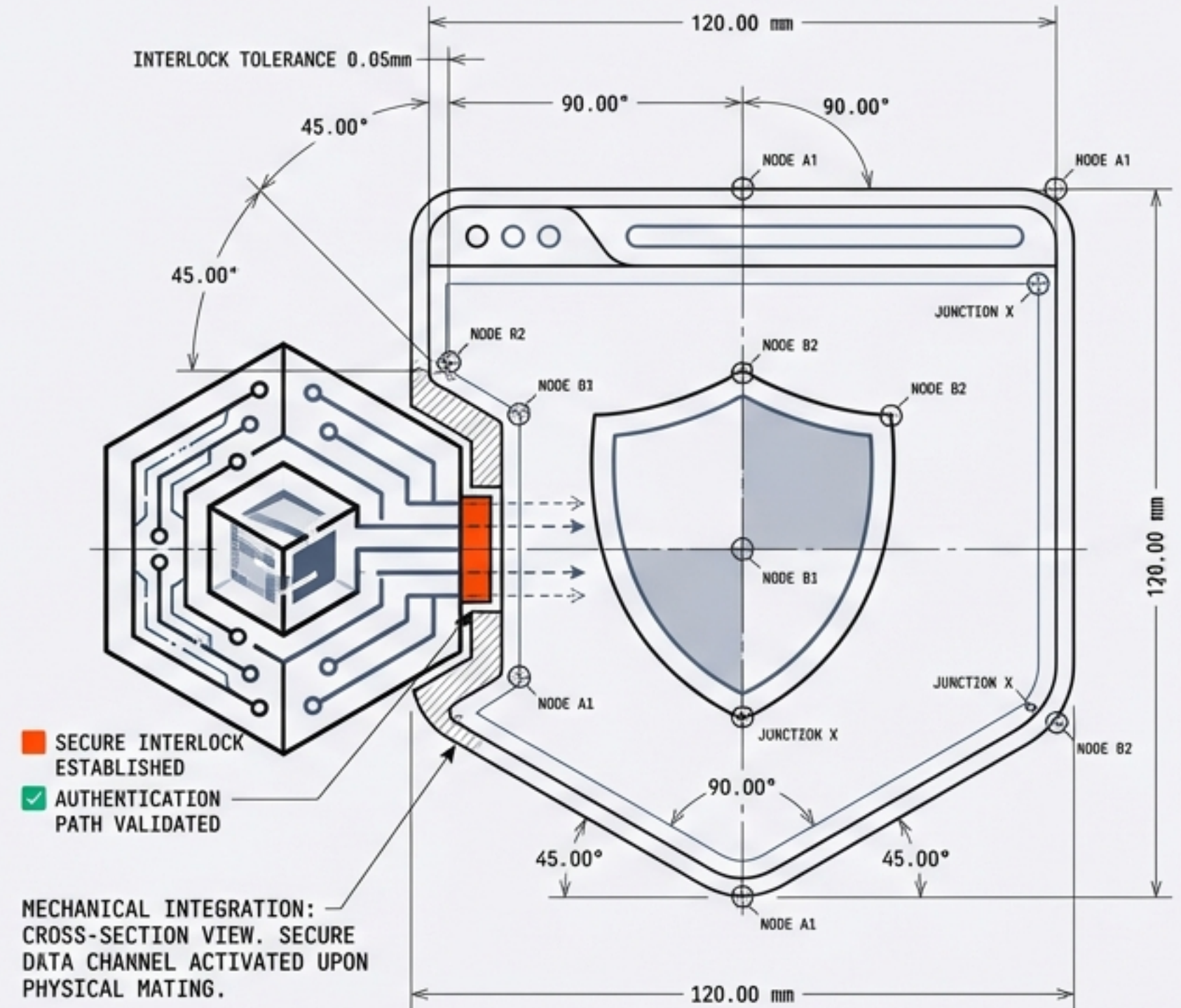


The Secure Browser Integration: Establishing Trusted Environments for SG/Send

Partnership Research Brief v0.4.11 |
Exploring synergies with Surf Security.

FROM: Human (Project Lead)
TO: Ambassador, Architect, AppSec
DATE: 19 Feb 2026

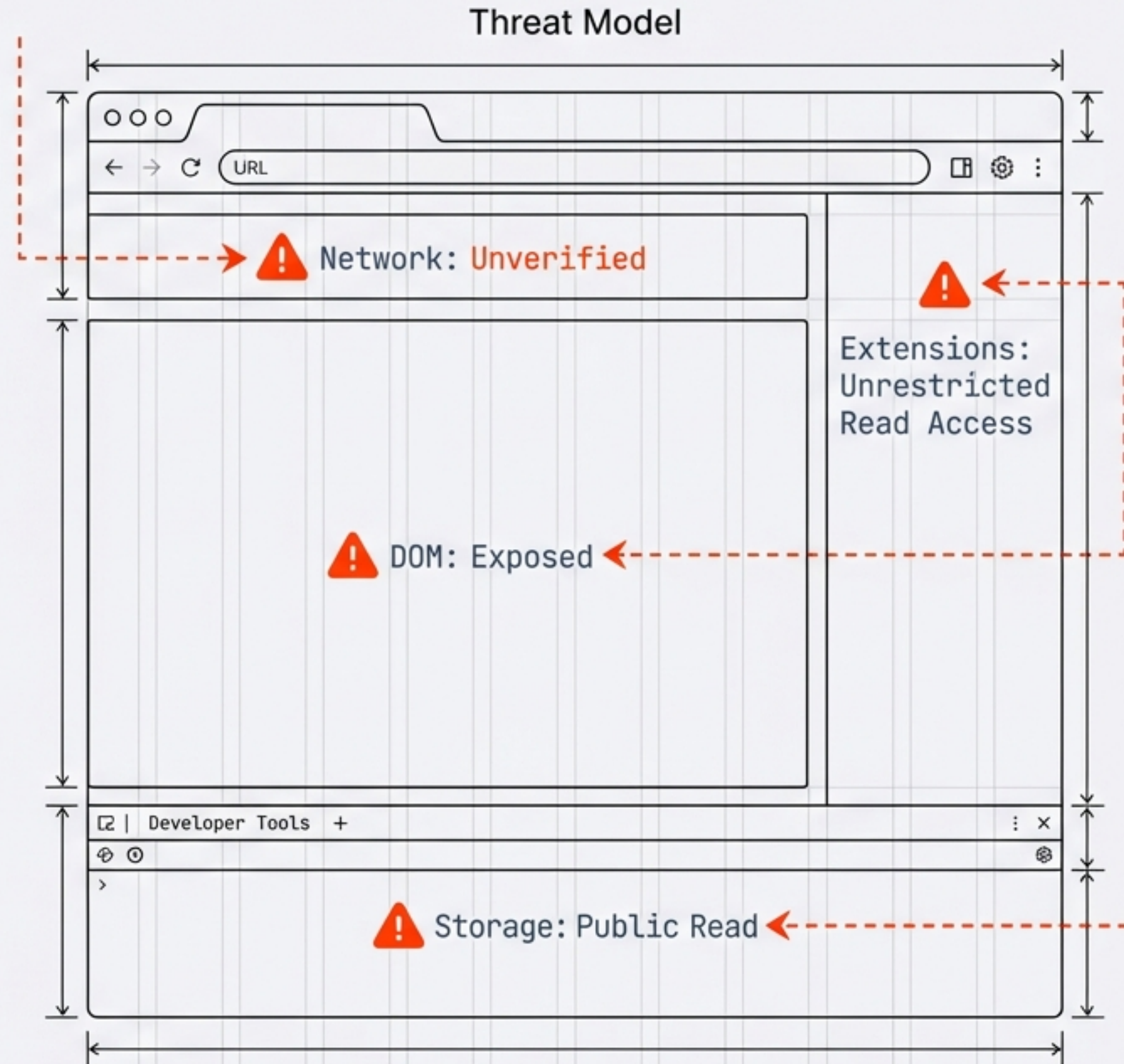


NOTE: While this brief uses Surf Security as the primary example due to existing relationships, the architectural concepts apply to the wider enterprise secure browser category (Island, Taton, etc.).

The Standard Browser is an Untrusted Environment.

We currently operate in a hostile environment where network integrity, extensions, and device posture are unverifiable variables.

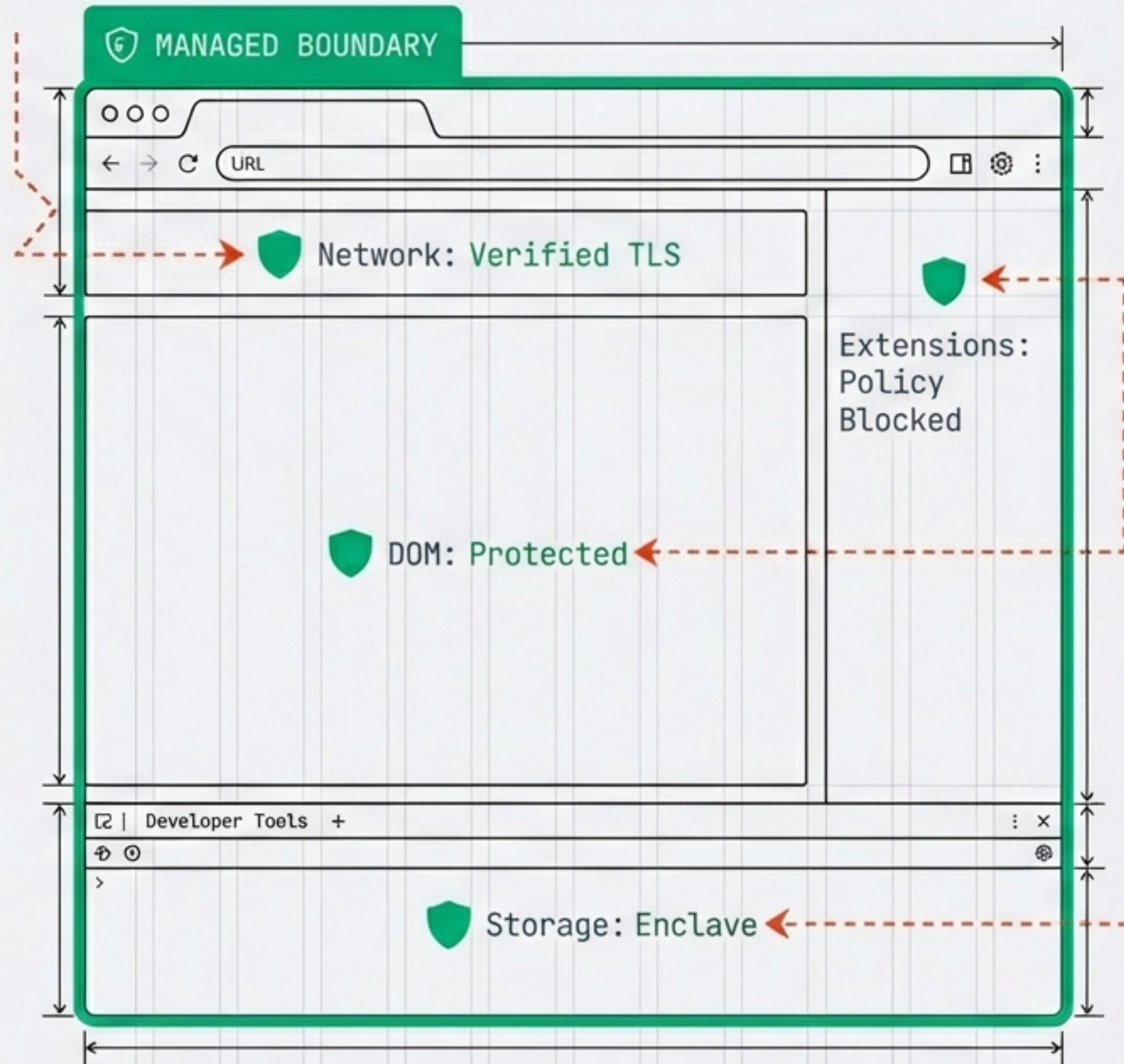
- ⚠️ **EXTENSION RISK:** Malicious extensions can read decrypted content directly from the DOM.
- ⚠️ **NETWORK RISK:** Unable to verify if traffic is being intercepted (MITM) or proxied.
- ⚠️ **DEVICE RISK:** Unknown if the host device is compromised or screen-recording.
- ⚠️ **STORAGE RISK:** Local storage is accessible to scripts with page access.



The Enterprise Browser Changes the Security Model.

Surf Security creates a managed boundary. The browser vendor and enterprise IT enforce policy, creating a “clean room” for execution.

- ✓ **EXTENSIONS:** Policy prevents unauthorized extensions from loading.
- ✓ **INTEGRITY:** Guaranteed TLS connection; no unauthorized proxies.
- ✓ **POSTURE:** The browser verifies device health before execution.



Achieving 'Impossible' Guarantees

"What if SG/Send running inside a secure browser could offer security guarantees that are mathematically impossible in a normal browser?"

**Zero-Knowledge
Encryption
(SG/Send) + Managed
Environment
(Surf) = Zero-Trust
Execution**

- Guarantee no extension reads the DOM.
- Prevent screen capture of decrypted content.
- True "End-to-Endpoint" security where the endpoint is the browser itself.

Integration Concept A: Browser-Managed Identity

Moving sensitive keys out of vulnerable storage.

Current State

Encryption keys reside in `localStorage`. Any script with page access can read them (a P3 finding from the security review).

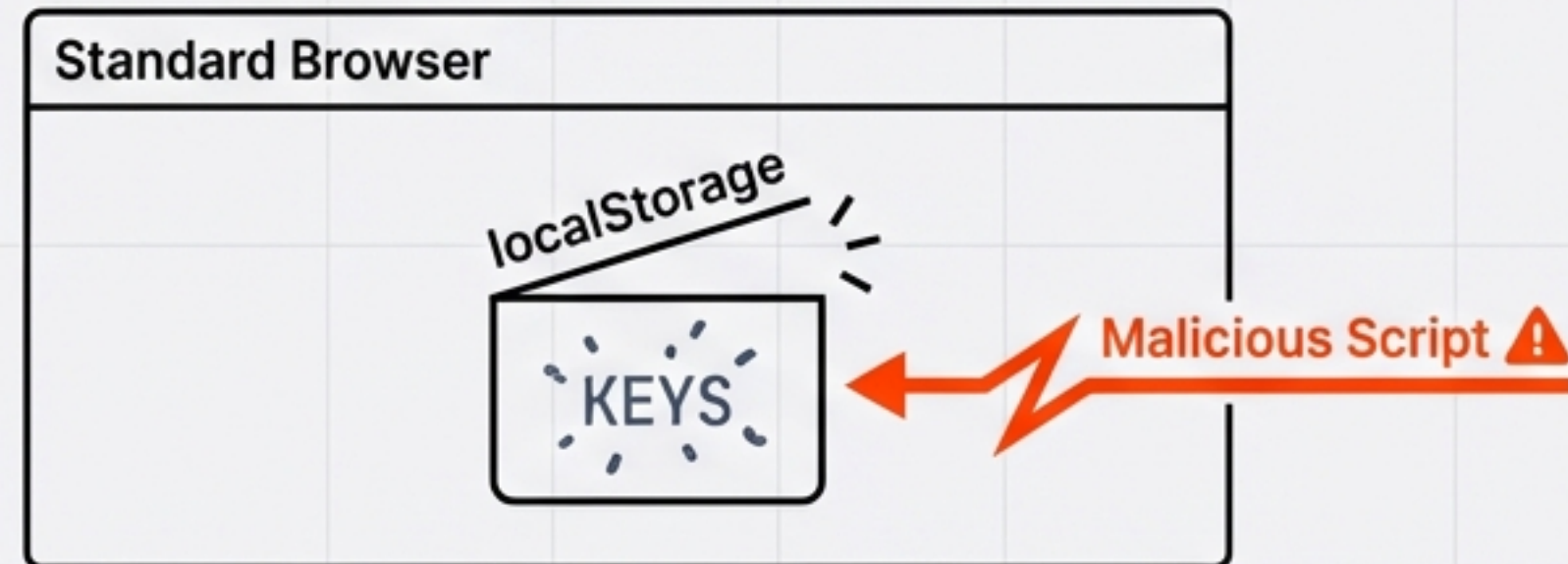
Future State

Secure Storage: Keys move to a browser-managed secure enclave or keychain.

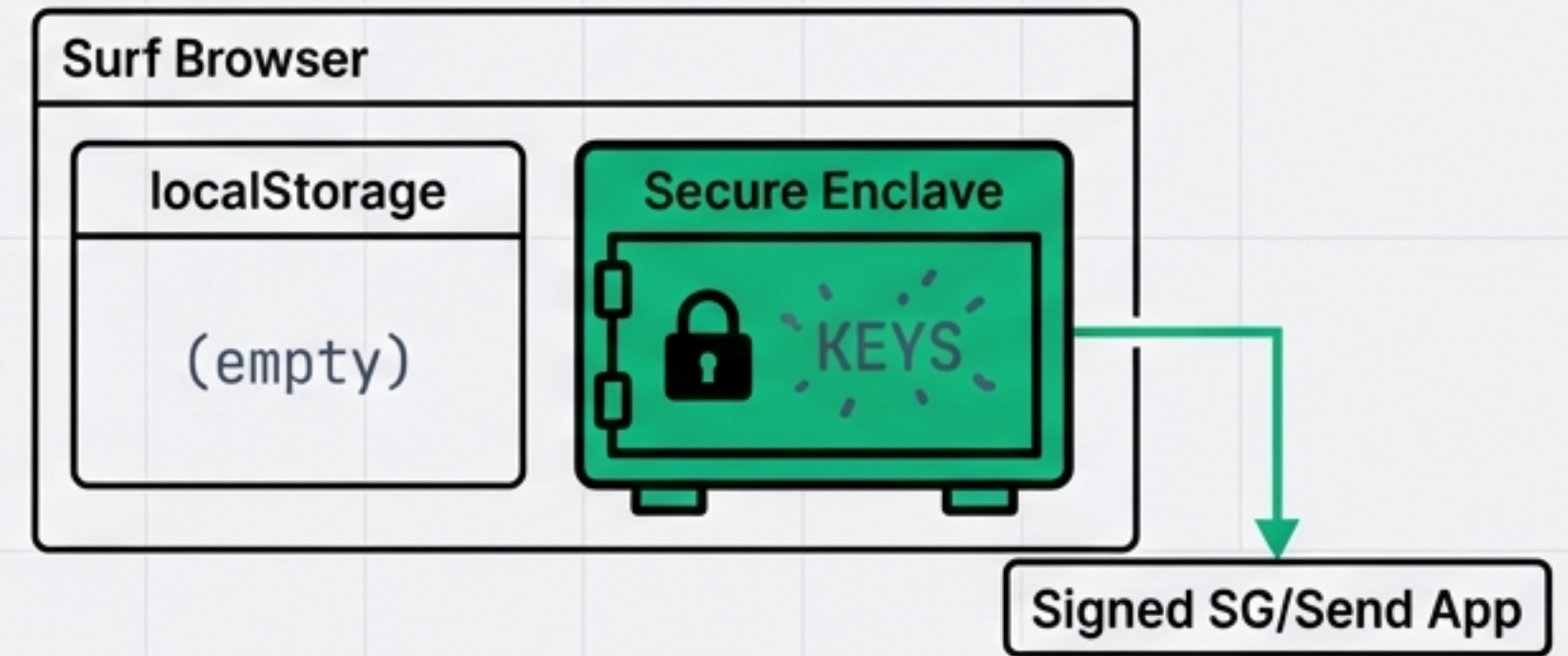
PKI Management: The browser manages the user's public/private key pair. Keys never leave the secure storage APIs.

STORAGE ARCHITECTURE

STANDARD BROWSER



SURF BROWSER



RESEARCH QUESTION: Does Surf (or similar) expose a protected storage API or certificate store that web apps can access via JavaScript?

Integration Concept B: Data Physics & DLP

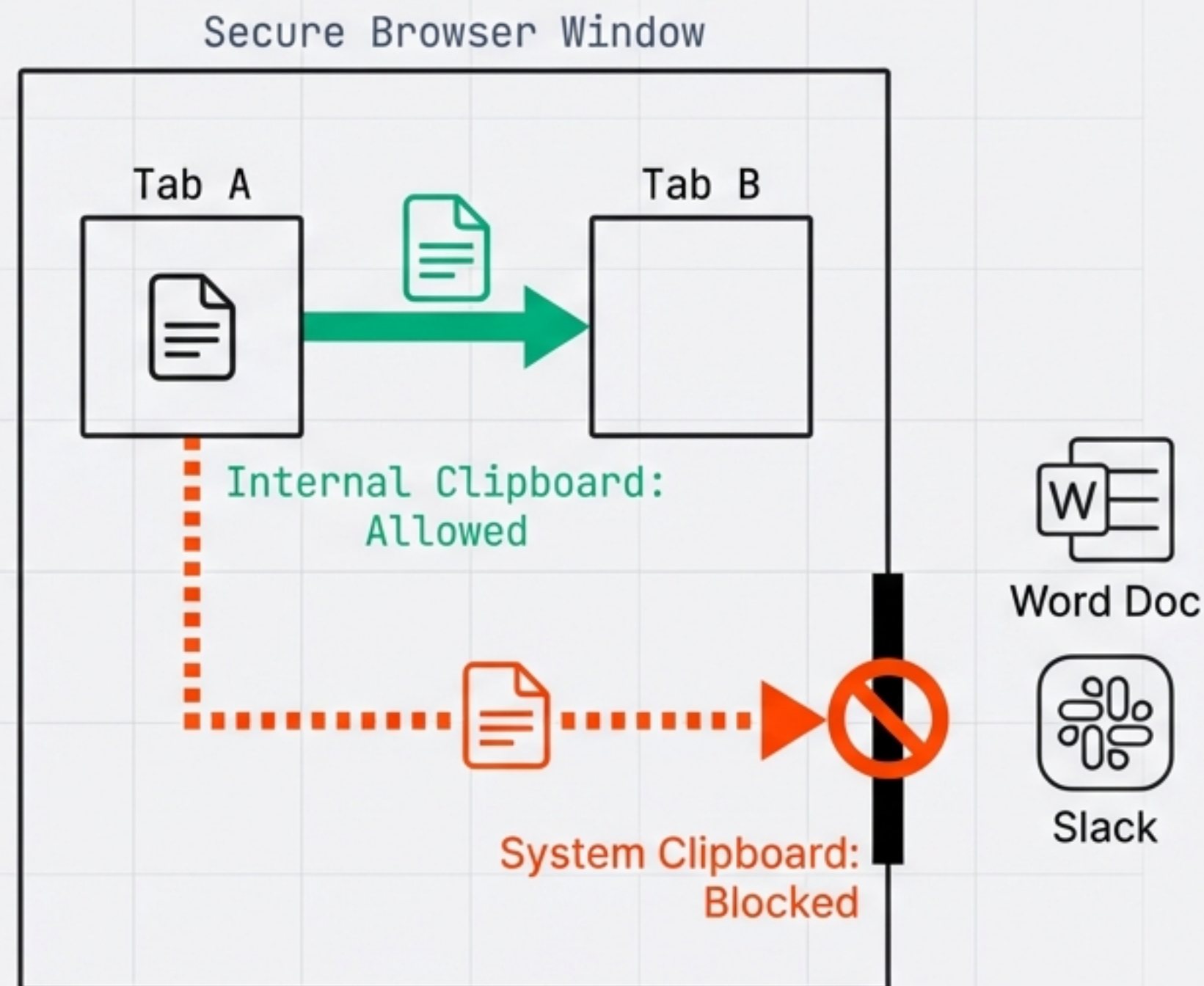
Enforcing “physics” on digital data. Content exists only in the tab and follows strict containment rules.

Idea 5: True View Once

- Recipient cannot save to disk.
- Recipient cannot screenshot (OS level block).
- Content is destroyed upon tab closure.

Idea 6: Secure Copy-Paste

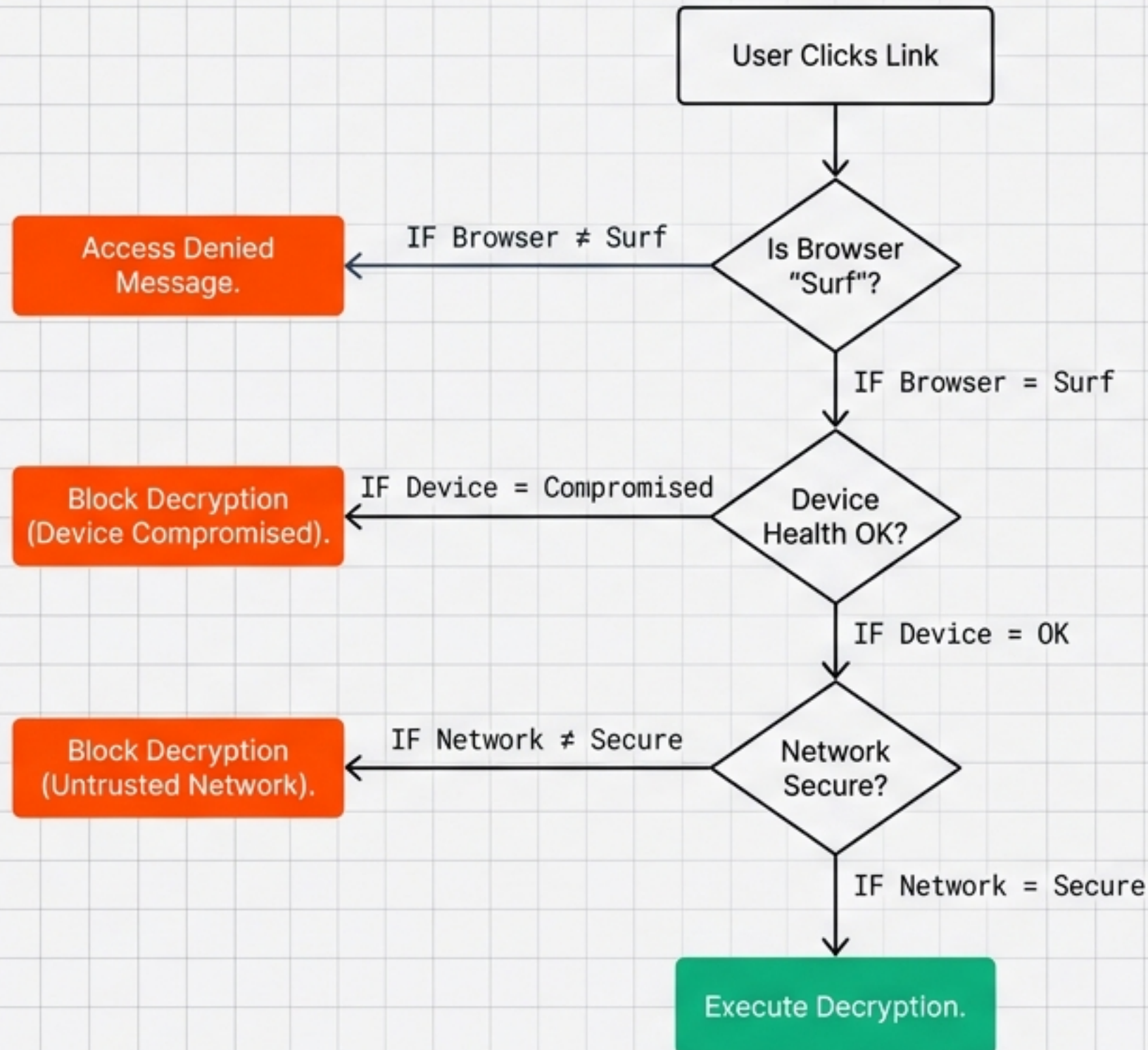
- Allowed: Copying from one SG/Send tab to another (internal workflow).
- Blocked: Copying from SG/Send to Word/Slack/External apps.



RESEARCH QUESTION: Can Surf enforce copy-paste and screenshot prevention policies specifically scoped to the SG/Send web application?

Integration Concept C: Contextual Access Control

Decryption is conditional based on the environment's integrity.



Strategic Value for SG/Send

Integration with an enterprise security vendor validates the SG/Send product and opens a direct channel to Surf's high-value enterprise customer base.

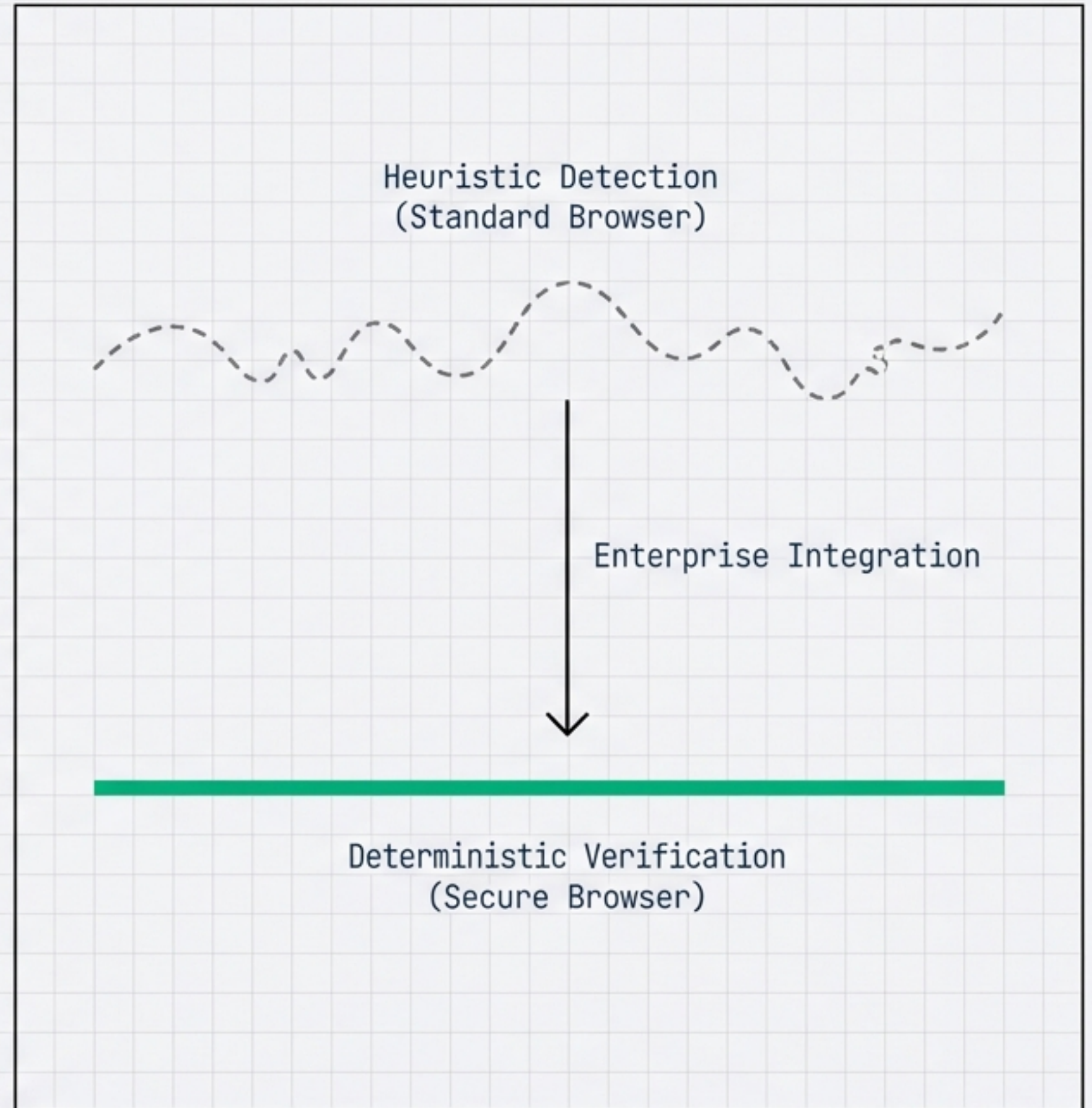
The "Solved" Problem

Reference: v0.4.7 Security Review (Chrome extension detection).

Resolution: For normal users, we use heuristics (guessing). For Enterprise/Surf users, we get definitive answers.

Revenue

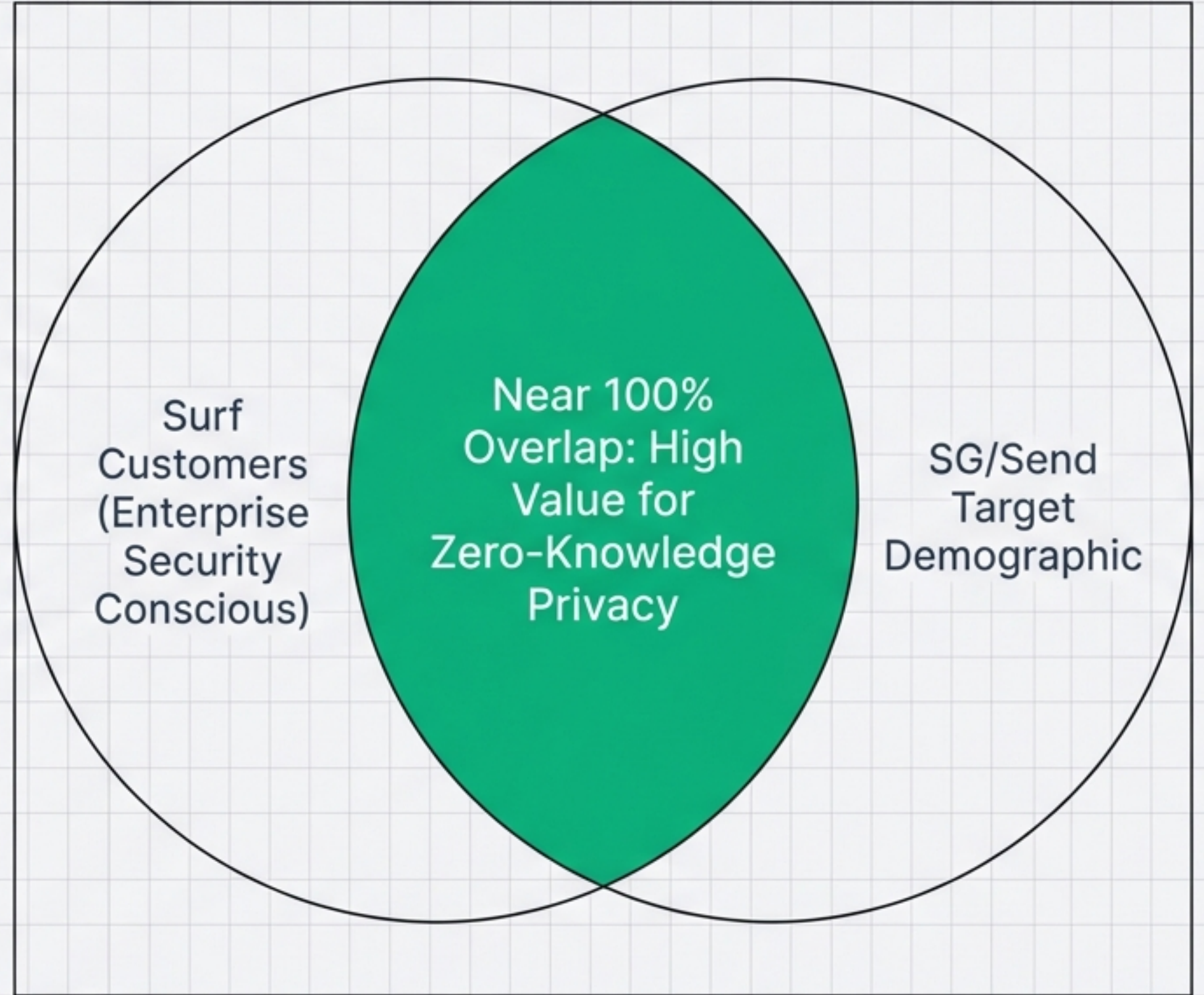
Potential for joint go-to-market and professional services.



Strategic Value for Surf Security

“Use Surf, and your SG/Send transfers become MORE secure.”

- **Differentiation:** Turns a passive tool (browser) into an active security enabler.
- **Stickiness:** Deepens the dependency on the managed browser for critical workflows.



Research Roadmap & Priorities

Validating technical feasibility before drafting the formal proposal.

TASK	DETAILS	OWNER	PRIORITY
API Capabilities	Research Surf Security's API. What can a web app detect?	Architect/Dev	P3
DLP Controls	Research anti-screenshot and copy-paste control capabilities.	AppSec	P3
Feasibility Study	Assess technical feasibility of the 6 integration ideas.	Architect	P3
Market Scope	Research Island and Talon APIs for comparison.	Architect	P4
Partnership Concept	Draft one-page partnership concept if research is positive.	Ambassador	P4

Redefining the Security Boundary

We are moving beyond simply encrypting the data. By controlling the execution environment, we bridge the gap between “Zero Knowledge” and “Zero Trust”. A shift from heuristic security (guessing) to deterministic security (knowing).

Data Protection
(Encryption)



Execution Protection
(Environment)

Shift from Heuristic (Guessing) → Deterministic (Knowing)